

# Rodrigo Alvarez Lucendo

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## EDUCATION

|           |                                                                                                                                                                                                                                                                                    |
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| 2023–2026 | <b>MSc Computer Science</b> , TU Delft <ul style="list-style-type: none"><li>Expected graduation: August 2026.</li><li>Exchange semester at EPFL (Feb–Jul 2025).</li><li>Courses: Modern NLP, Deep Learning, Graph ML, Deep RL, Computer Vision, Distributed ML Systems.</li></ul> |
| 2019–2023 | <b>BSc Computer Science</b> , TU Delft <ul style="list-style-type: none"><li>Thesis: Deep learning model for algal bloom forecasting using spatiotemporal data on HPC.</li></ul>                                                                                                   |

## WORK EXPERIENCE

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun–Present       | <b>Machine Learning Engineer</b> , Xcert.ai <ul style="list-style-type: none"><li>First employee at CHF 1.5M+ startup building AI assistants for aerospace certification.</li><li>Led RAG pipeline development (parsing, chunking, retrieval, and evaluation) as well as the development, monitoring and evaluation of an agentic workflow created with OpenAI Agent SDK.</li></ul> |
| Jul 2024–Jan 2025 | <b>Data Engineer Intern</b> , Amazon <ul style="list-style-type: none"><li>Deployed a monitoring tool for machine learning models in production.</li><li>First in team to explore AI agents; built a ReAct agent for natural-language metric analysis.</li><li>Provided BI teams with complex SQL queries and business analysis.</li></ul>                                          |
| 2022–2024         | <b>Teaching Assistant</b> , TU Delft & EPFL <ul style="list-style-type: none"><li>Guided 300+ students in BSc/MSc courses (ML, Big Data, Deep Learning); Head TA for Computational Intelligence.</li></ul>                                                                                                                                                                          |

## RESEARCH

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|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feb–Sep 2025 | <b>Research Assistant</b> , Theory of Machine Learning Lab, EPFL <ul style="list-style-type: none"><li>Investigated how Transformers learn high-order Markov chains through staged dynamics.</li></ul>                                                                         |
| 2025–2026    | <b>Master Thesis</b> , Natural Language Processing Lab, EPFL <ul style="list-style-type: none"><li>Measuring scaling laws for memorization in LLMs across model size and data composition.</li><li>Training GPT-2 models from scratch on synthetic and natural data.</li></ul> |

## PROJECTS

**AI Tutor:** Fine-tuned a 0.6B LLM with SFT and DPO (Hugging Face) to create an AI STEM tutor.

**nanoGPT:** Built nanoGPT from scratch (decoder-only Transformer, BPE tokenizer) and extended it with MoE.

**RAG QA:** Analyzed how contextual relevance influences RAG QA system performance.

**FETCH, EPFL AI Team:** Co-developed a natural-language mission planner for a Unitree Go2 robot using LLMs and live sensor input; integrated Nav2 and MCP servers.

## PUBLICATIONS

**Incremental Learning of Sparse Attention Patterns in Transformers.** Oğuz Kaan Yüksel, **Rodrigo Alvarez Lucendo**, Nicolas Flammarion. Second author; accepted to ICML 2026 Main Conference (regular accept), attending as author. Earlier version at EurIPS 2025 Workshop on Principles of Generative Modeling (PriGM). arXiv:2602.19143.

## SKILLS

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|------------------------|-------------------------------------------------------------------------|
| <b>Languages</b>       | Python, Java, SQL, Scala, Bash, TypeScript, English, Spanish.           |
| <b>ML / Deep Tech</b>  | PyTorch, Hugging Face, LlamaIndex, PyTorch Lightning, Weights & Biases. |
| <b>Systems / Tools</b> | Docker, AWS, Neo4j, Redis, FastAPI, Git, HPC.                           |
| <b>Interests</b>       | Rowing, alpine skiing, strength training.                               |